GlucoTwin — System Architecture

Patient-specific digital twin for Type 2 Diabetes that fuses EHR + wearable streams to forecast glucose spikes (>180 mg/dL) 2 hours ahead

1 · DATA STREAMS (SYNTHETIC, DPDP/HIPAA-SAFE)

Static / historical — EHR

Demographics, ICD-10 diagnoses (E11.9, R73.03, I10), LOINC labs (HbA1c, FPG, fasting insulin, lipids, eGFR), medications, genetic markers (TCF7L2, KCNQ1, FTO)

Synthea-style generator · FHIR R4 bundles

Dynamic / real-time — wearables

CGM glucose (5-min), heart rate, HRV (RMSSD), steps, sleep stages (W/L/D/R)

Apple Health / Google Fit / Libre-like schema

Patient app

Meal log (incomplete, noisy carb estimates)

Ingestion & interoperability

FHIR R4 adapter (ABDM-ready) · 5-min resampling · unit harmonisation · data-quality checks

Swap-in: MIMIC-IV, Synthea, OhioT1DM

2 · DIGITAL TWIN ENGINE (ONE VIRTUAL PATIENT PER PATIENT)

M Mechanistic core — extended Bergman minimal model

Glucose–insulin kinetics + 2-compartment gut absorption + exercise uptake; sleep modulates insulin sensitivity

$$dG/dt = -(S_G + X + E)\cdot G + S_G\cdot G_b + Ra/V \qquad dX/dt = -p_2X + p_2S_I(I - I_b)$$

$$dI/dt = -n(I - I_b) + \gamma\cdot(G - h)^+ \qquad Ra = f\cdot k_{abs}\cdot Q_2 \qquad dE/dt = -E/\tau + k_{ex}\cdot activity$$

1 Initialise

EHR → prior S_I (HOMA-IR, BMI), β-cell γ (diagnosis, TCF7L2/KCNQ1, sulfonylurea/DPP-4i), G_b (FPG)

2 Calibrate

3 onboarding days · MAP grid search on S_I, γ · robust Huber loss vs CGM · G_b from night CGM

3 Synchronise (every 5 min)

State observer pulls virtual G to CGM · unlogged-meal detector injects carbs into gut state

4 Forecast (every 15 min)

Run twin 2 h ahead: "no further meal" and "habitual routine" (learned meal-time profile)

5 What-if simulation

Counterfactuals from the live twin state: dinner carbs −25/−50%, post-meal walk 15/30 min, short sleep, missed metformin → 24-h glucose curve, peak, minutes >180, TIR

Hybrid ML risk layer — gradient-boosted trees (scikit-learn HistGradientBoosting)

58 fused features: CGM dynamics · HR/HRV/steps/sleep · meal log & time context · EHR · twin forecasts & personalised parameters

Output: P(CGM > 180 within 120 min) · alert threshold set on training patients · patient-level hold-out evaluation · ablation M1→M4 · group explanations (log-odds)

↻ continuous loop: new CGM / wearable sample → synchronise → forecast → risk

3 · CLINICIAN & PATIENT LAYER

Doctor dashboard (conceptual UI, working)

Patient list ranked by live risk · 72-h CGM + twin + risk timeline · alert explanation by data stream · twin physiology (prior vs calibrated) · what-if simulator · HR/HRV, steps, sleep

Alerts & nudges

"Spike likely in ~85 min" → suggested action (smaller portion, walk after meal) · escalation to care team

Outcomes tracked

Time-in-range, GMI, CV, spikes/day, false alerts per patient-day

Governance & safety

Synthetic data only · DPDP Act 2023 consent & purpose limitation · de-identification · human-in-the-loop (decision support, not autonomous) · model card & limitations · CDSCO SaMD pathway for clinical use

Tech stack

Python 3 · NumPy · pandas · SciPy · scikit-learn · Matplotlib · HTML/JS + Chart.js · FHIR R4 JSON · MIT license

Held-out results (30 unseen patients): AUROC 0.942 · 77% of spike episodes flagged · median 85-min warning · 0.79 false alerts/patient-day (4.6× fewer than CGM-only)